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A design method of micro-perforated panel absorber at high sound pressure environment in launcher fairings

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ABSTRACTS

A design method of micro-perforated panel absorbers is proposed for the enhancement of acoustic absorption inside the payload fairings of launch vehicles. An empirical acoustic resistance model of a micro-perforated panel absorber in a high sound pressure environment is suggested to this end. The resistance model is especially appropriate for the launcher fairing application, so that the incident pressure to the surface of bare fairings is considered as a main variable. Other design variables, e.g., the minute hole diameter, the perforation ratio and the thickness of the panel, are also considered in the proposed model. The estimated acoustic resistances and corresponding absorption coefficients are verified by comparing them with the measured results for several micro-perforated panel absorbers having a practical range of parameters. The estimated and measured results show good agreement when the perforation ratio is in the range of 1.4% to 5.3%. The effects of design parameters on the absorption coefficient are discussed. It is demonstrated that the micro-perforated panel absorber with a large hole diameter (about 1.0 mm), which shows a poor sound absorption at low pressure levels, can be a good sound absorber at high incident pressure levels. A set of design charts is provided and this enables one to readily design a micro-perforated panel absorber for the mitigation of acoustic loads in launcher fairings.

1. Introduction

The micro-perforated panel absorber (MPPA) [1] is one of the most promising noise control elements because of its applicability to extreme environments where general porous materials cannot be used. In particular, it does not contaminate the interior space as porous materials usually do by raising particles of dust. One of the applications of MPPAs has been in architectural acoustics [2–4]. When MPPAs are fabricated from metals, they can withstand high temperatures so that they can be applied to the reduction of aircraft engine noise [5].

Lift off acoustic loads generated by the propulsion systems of space launch vehicles can be one of the major threats to the survivability of payloads. Many commercial launchers have their own acoustic protection systems (APS) inside the payload fairings, e.g., acoustic blankets [6–8] and arrays of acoustic resonators [9–11]. One of the main design requirements of the acoustic protection system is strict cleanness (usually 10,000 to 100,000 class) to protect the payload from possible contamination. Since the MPPA is an inherent non-porous sound absorber, it can be a good candidate for the acoustic protection system of space launchers. The overall sound pressure level inside the payload fairings of commercial

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launch vehicles is so high (around 140 dB OASPL (overall sound pressure level)) that we cannot apply the linear impedance model [1] to the design of the acoustic protection systems directly. The high sound pressure level induces high particle velocity in the minute hole of micro-perforated panels. Ingard and Labate [12] showed the acoustic circulation in mid-intensity sound region. When the sound pressure increases further, they observed the formation of jets and vortex rings occurring at the exit of orifices. These effects increase the acoustic resistance significantly. This means that the excitation level of sound pressure must be one of the crucial parameters in designing MPPAs for launch vehicle applications.

The nonlinear impedance of micro-perforated panels (MPP) has been studied by a few researchers [13,14]. Maa [13] suggested the nonlinear acoustic impedance of MPP in terms of Mach number in the minute hole of MPP and the perforation ratio. Recently, Tayong et al. [14] proposed an improved nonlinear impedance model for MPP at high levels of excitation. The model was based on the linear equation of particle velocity in the perforation suggested by Bies et al. [15] and Ingard et al. [16].

When we design acoustic protection systems for launcher fairings, we usually have the frequency spectrum of sound pressure level in a band (usually in the 1/3 octave band) in bare fairings and the required sound pressure level (SPL) reduction. From the designer's point of view, the proper design variable is sound pressure level in the bare fairings. It would be preferable to have a simple impedance model expressed by the excitation sound pressure in the band of our interest rather than to have one based on the velocity inside the perforation.

In this paper, we propose an empirical nonlinear impedance model of a MPPA that can be readily applied to the design of the acoustic protection system. The proposed nonlinear resistance considers all geometric parameters, e.g., the perforation ratio, the diameter of the holes, the thickness of the panel as well as the excitation pressure. We incorporate the excitation pressure in the nonlinear impedance model by considering the relation between the incident pressure and the velocity inside an orifice [16]. We consider the geometric parameters in the form of non-dimensional terms by using Buckingham Pi theorem [17,18]. Their dependence on the nonlinear resistance is investigated by experiments for 27 MPPs whose dimensions are within the range of our interest. We verify the proposed resistance model of the MPPA experimentally. The current model helps one to estimate optimal dimensional parameters, which can maximize the absorption coefficient for high sound pressure levels inside the fairings of launch vehicles. It is demonstrated that the micro-perforated panel absorber with a large hole diameter (around 1.0 mm), which shows a poor sound absorption at low pressure levels, can be a good sound absorber in high sound pressure environments such as rocket fairings. The large hole diameter does not make the sound absorption bandwidth narrower at the high levels. This gives us a practical benefit in manufacturing MPPA by avoiding the fabrication of minute holes. The practical applicability of the proposed model is illustrated by the boundary element analysis of rocket fairings. It is shown that the application of the MPPA designed by the propose method could reduce noise level by more than 6 dB inside the fairings.

2. An empirical acoustic impedance model for high sound pressure environment

2.1. Brief review of the linear acoustic impedance model of MPPA

The normalized specific acoustic impedance of a micro-perforated panel absorber (MPPA) can be written as

$$z_l = r_l + jx_l \tag{1}$$

where r_l is the normalized specific acoustic resistance and x_l is the normalized specific acoustic reactance. Each term can be expressed as [1]

$$r_l = \frac{32\eta t}{\rho_0 c_0 d^2 \sigma}\left(\sqrt{1+\frac{k_p^2}{32}}+\frac{\sqrt{2}}{32}k_p\frac{d}{t}\right), \tag{2}$$

$$x_l = \frac{\omega}{c_0}\frac{t}{\sigma}\left(1+\frac{1}{\sqrt{9+k_p^2/2}}+0.85\frac{d}{t}\right)-\cot\left(\frac{\omega D}{c_0}\right) \tag{3}$$

This model is valid when $1<k_p<10$ [19]. In the above equations, ρ_0 is the density of air, η is the coefficient of kinematic viscosity of air, c_0 is the speed of sound in air, ω is the frequency in radian, t is the thickness of the MPP, d is the diameter of the minute hole, and σ is the perforation ratio. The perforation ratio is defined as the total perforated area divided by the area of the MPP. The depth of the back cavity is denoted by D. Note that k_p is the quantity that is proportional to the ratio of the radius to the viscous boundary layer thickness inside the orifice and this can be defined as $k_p = 0.5d\sqrt{\rho_0\omega/\eta}$. When the hole diameter becomes smaller (below 1 mm in diameter), the relative viscous boundary layer thickness on the inner surface of the hole becomes larger. This increases the acoustic resistance and makes the MPPA wide band sound absorbers. It is known that the above model is valid when the excitation sound pressure level is low (below 100 dB [1]).

If one increases the excitation, the sound pressure level of the flow in the minute hole separates at the exit of the orifice. This causes nonlinear phenomena [12], which dissipate acoustic energy. Thus, the resistance of the MPPA increases significantly in the high pressure environment. The increase of the resistance changes the peak of absorption coefficient.

The high excitation pressure also changes the acoustic reactance due to the nonlinear flow phenomena. The decreased acoustic end correction length makes the peak frequency of the absorption move to high-frequency region. However, the degree of frequency shift is so small that it does not affect the absorption characteristics significantly. In this paper, we focus on the nonlinear resistance of the MPPA rather than the reactance. We can apply the conventional nonlinear reactance model [13] to the design.

2.2. Simplification of the problem

In practice, the sound field inside launcher fairings can be regarded as a diffuse field. This can be decomposed into the sum of plane waves (by the spatial Fourier transform) in all possible directions. It is noteworthy that one can regard the MPPA as a locally reacting absorber where sound propagation is not possible in the parallel direction to the MPP surface. The locally reacting absorber is characterized by wall impedance that is independent of the angle of incidence [20]: The normal impedance can be a sufficient measure to characterize the acoustic property of the MPPA. (This can be readily measured in the impedance tube.) In this regard, the real three-dimensional situation can be simplified to the one-dimensional situation without loss of generality. This helps us to focus on the normal impedance model of the MPPA to deal with the problem. For example, if one has the normal impedance model of the MPPA, one can readily obtain the boundary impedance condition in the boundary integral equation, which enables one to estimate the noise reduction by applying the MPPA to the fairings of launch vehicles. In addition, one can use the normal incident absorption coefficient to express the sound absorbing ability of the MPPA.

It is worth noting that the use of the total pressure on the surface of the MPPA as a primary control variable is not feasible. It is related to not only the blocked pressure but also the radiation impedance and the volume velocity of the porous structure of the absorber [21]. The value of the total pressure depends on the existence of sound absorbers on the fairing surface. Instead of using it, we made use of the incident pressure that is invariable regardless of the impedance of the boundary surface. When we design the MPPA for the mitigation of noise inside the fairings, we can regard the specified acoustic loading on the bare fairing surface as twice the incident pressure (see Fig. 1(a)). Thus, we define the incident pressure as the primary variable of the proposed resistance model. In the simplified approach, the incident pressure level can be defined by measuring the blocked pressure level on the surface of a composite sandwich plate that mimics the bare surface of the launcher fairings. After we replace the composite sandwich plate with a MPPA, we can measure the impedance according to the specified incident pressure level. Fig. 1(b) illustrates this procedure.

2.3. Experimental procedure

We manufactured a total of 27 micro-perforated panels whose common size is 130 mm $\times$ 130 mm to investigate the nonlinear acoustic impedance of MPPAs. The panels are made of light carbon fabric reinforced plastic (CFRP) materials considering both the reduction of weight and increase in stiffness for the application to launch vehicles. We made 18

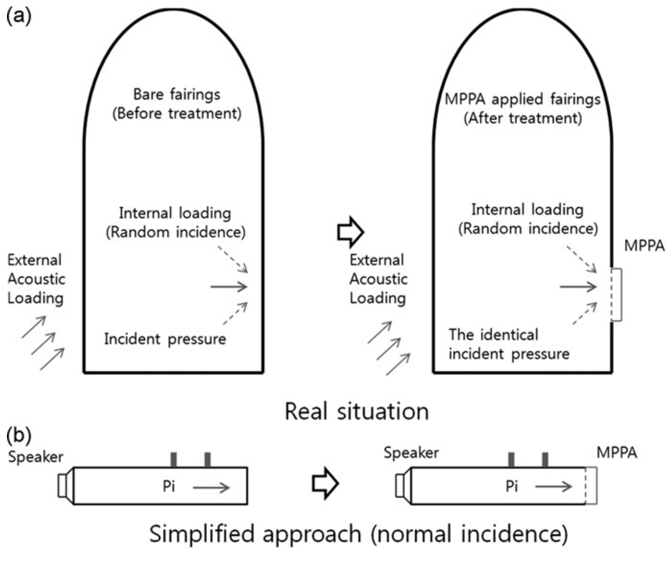

Fig. 1. Simplification of real situation to one-dimensional normal incidence case for an experimental investigation. (a) The internal acoustic loading on the bare fairing surface can be regarded as twice the incident pressure. The incident pressure can be assumed to be invariable regardless of the surface boundary condition. (b) The real situation can be simplified to one-dimensional case in the impedance tube. The measure normal impedance can be used to predict the internal pressure field in the fairings by the boundary integral equation (or boundary element method) since the MPPA can be regarded as a locally reacting sound absorber.

micro-perforated panels of 1.0 mm thickness and 9 panels of 2.0 mm thickness. The range of the minute hole diameters is 0.4 mm to 1.0 mm, and the range of perforation ratios is 0.6% to 5.7%. Fig. 2 shows one of the MPPs for the experiments. The depths of the cavities of the MPPAs range from 70 mm to 100 mm considering the dimensional restrictions of the acoustic protection system for payload fairings. Notice that the maximum height of the conventional acoustic protection systems is below 100 mm to provide enough payload volume. We attached a horn driver to the one end of the impedance tube whose cross-sectional area is $90\times 90\ \text{mm}^2$ to generate band-limited random noise. We recorded the relation between the input voltage level to a power amplifier and the corresponding incident pressure level when we attached the sample of the launcher fairing skin on the other end of the duct. This recorded input voltage level was used to excite all MPPAs to measure the acoustic impedance variation according to the incident pressure level. The acoustic impedance of the MPPA can be estimated by the two microphone method [22]. It is observed that the horn driver excited the MPPAs evenly in the 1/3 octave bands whose center frequency ranges from 200 Hz to 630 Hz where the absorption coefficient peaks exist. The difference of the pressure levels between each 1/3 octave band falls within 2 dB. The maximum incident pressure level is about 143 dB overall (the sum of the 200 Hz, 250 Hz, 315 Hz, 400 Hz, 500 Hz, and 630 Hz-1/3 octave bands), which is sufficient to simulate the internal acoustic loading level inside the payload fairings of the launchers. We measured all acoustic impedances for nine different incident pressure levels (100 dB, 103 dB, 109 dB, 115 dB, 121 dB, 127 dB, 133 dB, 139 dB, and 143 dB) for each MPPA to investigate the effect of the incident pressure levels on the impedance.

2.4. Acoustic resistance model of MPPA by dimensional analysis

The experimental results indicate that the acoustic resistance increases as the excitation level increases. We also observed that the acoustic reactance curves shifts to the high frequency region (this means that the acoustic inertance decreases due to the high particle velocity in the perforate), but the amount of shift does not significantly alter the acoustic absorption coefficient characteristics. These have been reported by other researchers [1,13,23].

At very high sound pressure levels, the nonlinear normalized specific resistance of the panel with an orifice can be expressed as $r=u_0/c_0\times(1-\sigma^2)/\sigma$ (u_0: the velocity in the orifice), which can be derived by Bernoulli's law [24]. When the perforation ratio is low, this becomes $r=u_0/c_0\sigma$. (Notice that the nonlinear part of the resistance is independent of other geometric parameters.) The nonlinear term in Maa's resistance model [13] ($r=r_l+u_0/c_0\sigma$) corresponds to this term. However, in our case, there is a large discrepancy between the estimated resistances by Maa's model and the measured values. The conventional model is not appropriate to describe the behavior of the measured nonlinear resistance thoroughly. It is desirable to have a new resistance model whose main control variable is the sound pressure, to facilitate the practical design of MPPAs.

An appropriate model of the nonlinear resistance of MPPs can be expressed as the sum of the linear term and the additional nonlinear term due to the high sound pressure,

$$R_{nl}=R_l+\Delta R_{nl} \tag{4}$$

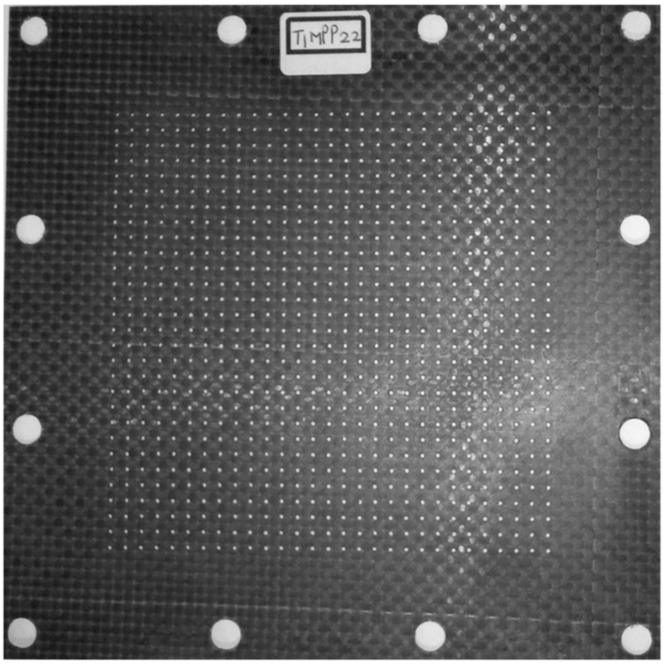

Fig. 2. Photo of a CFRP MPP (thickness = 1.0 mm, diameter = 0.8 mm, perforation ratio = 5.3%, size: $130\times 130\ \text{mm}^2$).

where R_{nl} is the total specific acoustic resistance, R_l is the specific acoustic resistance at sufficiently low pressure levels (Eq. (2) multiplied by $\rho_0 c_0$), and ΔR_{nl} is the nonlinear part of the resistance. We carry out dimensional analysis [18] to determine non-dimensional variables that govern the nonlinear part of the resistance (ΔR_{nl}). We assume that ΔR_{nl} is a function of the density of air (ρ_0), the diameter of the hole (d), the thickness of the MPP (t), the distance between adjacent holes (b), the root-mean-squared (rms) velocity in the hole ($\overline{u}_0$), and the speed of sound (c_0), then, we may express the relationship among the variables in functional form as

$$\Delta R_{nl} = f(\rho_0, d, t, b, \overline{u}_0, c_0). \tag{5}$$

In Eq. (5), the nonlinear part of the resistance is assumed independent of the frequency. The repeating parameters, which are equal to the number of fundamental dimensions, are chosen as ρ_0, d, $\overline{u}_0$. Buckingham Pi theorem [18] results in four dimensionless groups. We can write the non-dimensional form of Eq. (5) as

$$\frac{\Delta R_{nl}}{\rho_0 c_0} = f\left(\frac{t}{d}, \frac{b}{d}, \frac{\overline{u}_0}{c_0}\right), \tag{6}$$

and in its simplest form, it may be represented as

$$\Delta r_{nl} = C_0 \left(\frac{d}{t}\right)^m \sigma^n \left(\frac{\overline{u}_0}{c_0}\right)^l, \tag{7}$$

where d/b is replaced by the perforation ratio ($\sigma = \pi d^2/4b^2$). The nonlinear part can be modeled as the product of the non-dimensional terms related to the dimensional variables and the term associated with the velocity in the hole. Previous studies [13,14,16,25] showed that the nonlinear resistance depends on the velocity ($l=1$). Note that Eq. (7) could be reduced to the nonlinear resistance model by Maa ($r = r_l + u_0/c_0\sigma$) when $C_0=1$, $m=0$, and $n=-1$.

It is very desirable to express the acoustic resistance in terms of the sound pressure level (SPL) in the band of interest because the acoustic loading inside the fairings is often specified by the SPL in the band. The rms velocity $\overline{u}_0$ in the hole can be expressed in terms of the rms incident pressure ($\overline{p}_i$), that is,

$$\frac{\overline{u}_0}{c_0} = \frac{1}{\sqrt{2}} \frac{\sigma}{1-\sigma^2} \left(\sqrt{0.25 + \frac{2\sqrt{2}\overline{p}_i}{\rho_0 c_0^2} \frac{1-\sigma^2}{\sigma^2}} - 0.5\right). \tag{8}$$

Note that this equation can be obtained by using the acoustic circuit analogy and Bernoulli's law [2,16,24]. The detailed derivation is shown in the Appendix. When the excitation is band-limited, the mean squared values ($\overline{p}_i^{\,2}$) are equivalent to the overall SPL in the frequency band (this can be verified by Parseval's theorem). The sound pressure level in the band (L_{Pi}) can be expressed as

$$L_{Pi} = 10 \log \frac{\overline{p}_i^2}{p_{\text{ref}}^{\,2}} (\text{dB}) \quad (p_{\text{ref}} = 20\ \mu\text{Pa}). \tag{9}$$

Fig. 3 illustrates that the relation between the rms blocked pressure ($2\overline{p}_i$) and the rms flow velocity in the orifice ($\overline{u}_0$) estimated by Eq. (8). The $2\overline{p}_i \sim \overline{u}_0$ relations are shown for 0.5%, 1.0%, 2.0%, and 5.0% perforation ratios. Ingard and Ising [16] observed the relation by the hot wire measurement when the perforation ratio of the orifice panel is about 0.5%. Fig. 3 demonstrates that the estimated results for 0.5% perforation ratio (grey solid line in Fig. 3) shows reasonably good agreement with the measured ones by Ingard and Ising (circles in Fig. 3) for $\overline{u}_0 > 5$ m/s. When the perforation ratio becomes higher, the

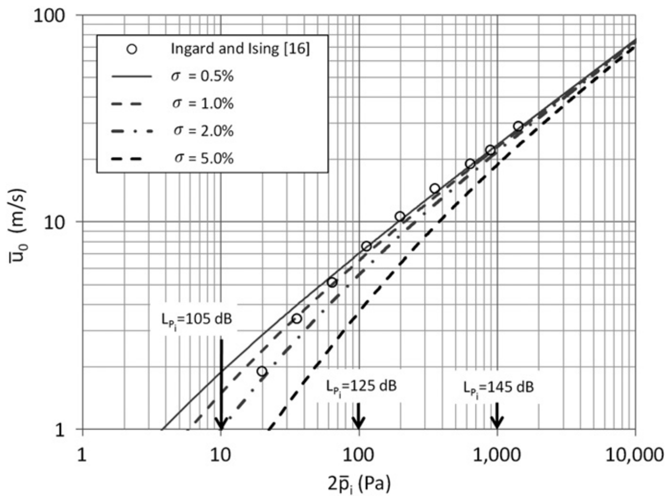

Fig. 3. The relation between the blocked pressure ($2\overline{p}_i$) and the flow velocity in the orifice ($\overline{u}_0$) estimated by Eq. (8). The estimated result is in good agreement with the measured one [16] at $\sigma=0.5\%$ for $\overline{u}_0 > 5$ m/s.

corresponding velocity in the hole becomes lower at the same incident pressure. This explains the small increment of the nonlinear resistance of high perforation ratio MPPs at increased pressure levels. The sound pressure level of our interest is in the range of 130 to 140 dB. In this SPL range, Eq. (8) may be a good estimate of the velocity in the orifice for a given level of the incident pressure. Eqs. (8) and (9) enable one to express the nonlinear resistance model in terms of the incident sound pressure level. By using Eqs. (7) and (8), the total normalized specific acoustic resistance (Eq. (4) divided by $\rho_0 c_0$) is expressed as

$$r_{nl}=r_l+\Delta r_{nl}=r_l+C\left(\frac{d}{t}\right)^m \sigma^n\left[\frac{\sigma}{1-\sigma^2}\left(\sqrt{0.25+\frac{2\sqrt{2}\overline{p}_i}{\rho_0 c_0^2}\frac{1-\sigma^2}{\sigma^2}}-0.5\right)\right]. \tag{10a}$$

where $C=C_0/\sqrt{2}$. For the low perforation ratio ($\sigma \ll 1$) MPPs, Eq. (10a) is reduced to

$$r_{nl}=r_l+C\left(\frac{d}{t}\right)^m \sigma^n\left[\sigma\left(\sqrt{0.25+\frac{2\sqrt{2}\overline{p}_i}{\rho_0 c_0^2 \sigma^2}}-0.5\right)\right]. \tag{10b}$$

where the incident SPL (L_{Pi}) is given, $\overline{p}_i$ is calculated by the relation, $\overline{p}_i=p_{ref}10^{L_{Pi}/10}$.

We use the averaged normalized specific acoustic resistance in six 1/3 octave bands (from the 200 to the 630 Hz-1/3 octave band) to determine C, m and n in Eq. (10). In order to determine the exponents m and n, the nonlinear part of the normalized resistance divided by the term associated with the incident pressure,$\Delta r_{nl}/(\sqrt{0.25+2\sqrt{2}\overline{p}_i/\rho_0 c_0^2\sigma^2}-0.5)$, are plotted according to the two non-dimensional variables, d/t and σ. (See Figs. 4 and 5.) One can determine the exponents m and n by the linear least-squares fitting method in the log–log plots. Fig. 4 illustrates that the exponent m is 0.06. Fig. 5 indicates that the exponent n is -0.845 ($n+1=0.155$ in Fig. 5). The constant C is found to be 1.59. Now the nonlinear normalized resistance of the MPPs is written as follows:

$$r_{nl}=r_l+1.59\left(\frac{d}{t}\right)^{0.06}\sigma^{-0.845}\left[\sigma\left(\sqrt{0.25+\frac{2\sqrt{2}\overline{p}_i}{\rho_0 c_0^2\sigma^2}}-0.5\right)\right]. \tag{11}$$

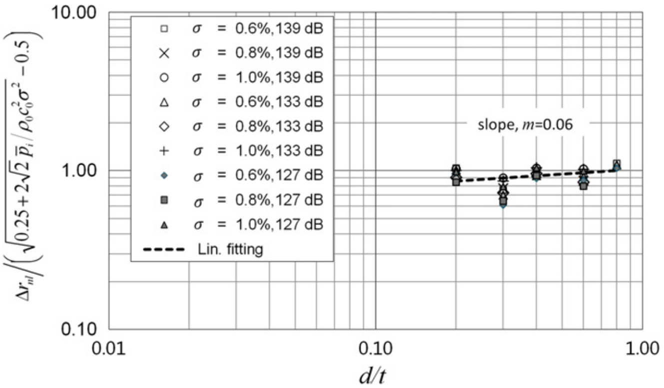

Fig. 4. The exponent m can be determined by dividing the nonlinear part of the normalized resistance by the term representing the incident pressure. The linear least–squares fitting method reveals $m=0.06$. Notice that the effect of d/t is not dominant in the nonlinear part of the resistance.

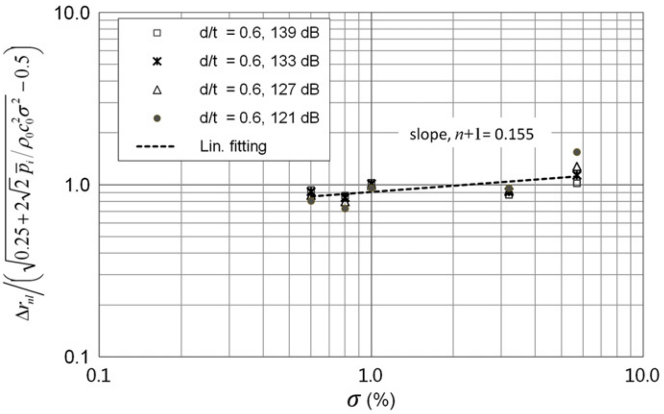

Fig. 5. The exponent n can be determined by dividing the nonlinear part of the normalized resistance by the term representing incident pressure. The linear least-squares fitting method reveals $n+1=0.155$ ($n=-0.845$).

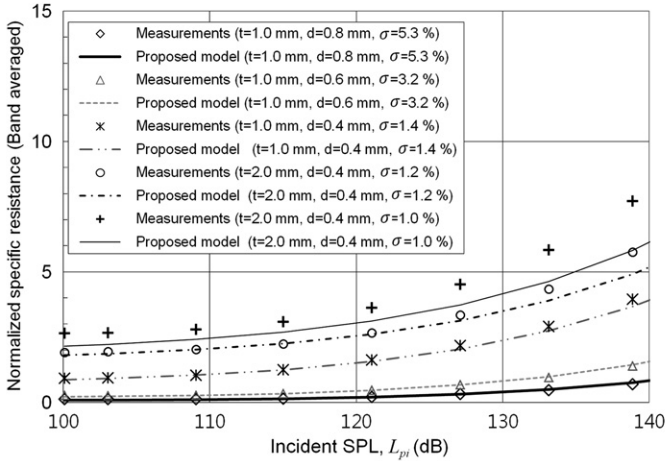

Fig. 6. Comparisons of measured normalized specific resistances and estimated ones by the proposed model. In the range of practical application, the maximum relative error is within $\pm 10\%$ at 143 dB ($\sigma=1.4$, 3.2, and 5.3%). The proposed model underestimates the resistance of MPPs whose perforation ratio is lower than 1.2%. The relative error is about -16% for the $\sigma=1.2\%$ MPP, it increases to -25% when the perforation ratio is very low ($\sigma=1.0\%$).

It is worth noting that the nonlinear part of the resistance is weakly dependent on the hole diameter and the thickness. This enables us to use a large-diameter hole in the design of MPPAs in high pressure levels, which often make MPPAs poor sound absorbers at low pressure levels. This will be demonstrated in the next chapter.

In order to verify the proposed resistance model, we compare the estimated nonlinear normalized specific resistances by Eq. (11) with the corresponding measured ones in Fig. 6. We make use of Eq. (11) for the estimation of the acoustic resistance. The results demonstrate that predicted resistances are in good agreement with the measured ones. The maximum relative error is within $\pm 10\%$ at 143 dB for the MPPs whose perforation ratios are 1.4, 3.2, and 5.3%, respectively. Since the sound pressure level in general rocket fairings is around 140 dB and these perforation ratios cover the practical range of MPPAs, Eq. (11) can be used to estimate the resistance of the MPPA for the rocket fairings. Eq. (11) underestimates the resistance of MPPs whose perforation ratio is lower than 1.2%. For the $\sigma=1.2\%$ MPP, the relative error becomes -16% at 143 dB. When the perforation ratio is very low ($\sigma=1.0\%$), the relative error increases to -25% at 143 dB. One has to bear the limitation of the proposed formula in mind when the perforation ratio of a MPP is very low.

We did not investigate the acoustic reactance in this paper. The conventional model can be applied to the design of MPPAs. According to Maa's model [13], the acoustic reactance of a MPPA at high SPL can be expressed as

$$x_{nl} = \frac{\omega}{c_0}\frac{t}{\sigma}\left(1+\frac{1}{\sqrt{9+k_p^2/2}}+0.85\frac{d}{t}\left(1+\frac{\sqrt{2}\overline{u}_0}{\sigma c_0}\right)^{-1}\right)-\cot\left(\frac{\omega D}{c_0}\right). \tag{12}$$

Eqs. (11) and (12) lead us to have the normalized specific acoustic impedance of the MPPA at high sound pressure level as follows:

$$z_{nl} = r_{nl}+jx_{nl}. \tag{13}$$

3. Verification examples of the nonlinear acoustic resistance model

We compare the estimated normal incidence absorption coefficient and the measured ones for three MPPAs. The geometric parameters of the MPPAs are summarized in Table 1. We performed an experiment for MPPA #1 whose thickness is 1.0 mm, the hole diameter is 1.0 mm and the perforation ratio is 5.14%. The depth of the back cavity is 100 mm. We measured the acoustic impedance at 115 dB and 143 dB (both in the overall SPL from the 200 Hz- to the 630 Hz-1/3 octave bands) by using the impedance tube. The corresponding impedances are predicted by using Eqs. (11) and (12) at the same pressure level. The normal incidence absorption coefficients are shown in Fig. 7. The estimated results by the present model and the measured ones are in good agreement in both low and high incident pressure levels. The results are consistent with those of Fig. 6. The linear resistance model (Eq. (2)) underestimates the absorption coefficient because this does not consider the resistance increment due to the high excitation pressure.

In Fig. 8, we compare the measured and predicted normal incidence absorption coefficients for MPPA #2 and MPPA #3. These two MPPAs have the same geometric parameters (the hole diameter is 0.4 mm and the perforation ratio is 1.4%) except for the thickness. The thickness of MPPA #2 is 1.0 mm and that of MPPA #3 is 2.0 mm. The depth of the back cavity of is 70 mm. We measured the absorption coefficient at 121 dB and predicted it at the same SPL by using Eq. (11). One can readily notice that the absorption coefficients of MPPAs #2 and #3 are different with each other. The thicker MPPA has the

Table 1
Geometric parameters of MPPAs for the verification of the proposed resistance model.

	Thickness (mm)	Hole diameter (mm)	Perforation ratio (%)	Cavity depth (mm)
MPPA #1	1.0	1.0	5.14	100
MPPA #2	1.0	0.4	1.4	70
MPPA #3	2.0	0.4	1.4	70

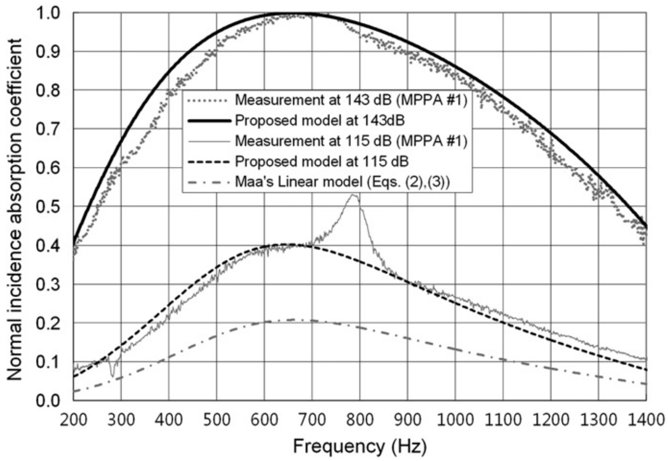

Fig. 7. Normal incidence absorption coefficient comparison of measured and estimated results of MPPA #1 (thickness = 1.0 mm, hole diameter = 1.0 mm, perforation ratio = 5.14%, depth of cavity = 100 mm). The results are estimated at 115 dB and 143 dB.

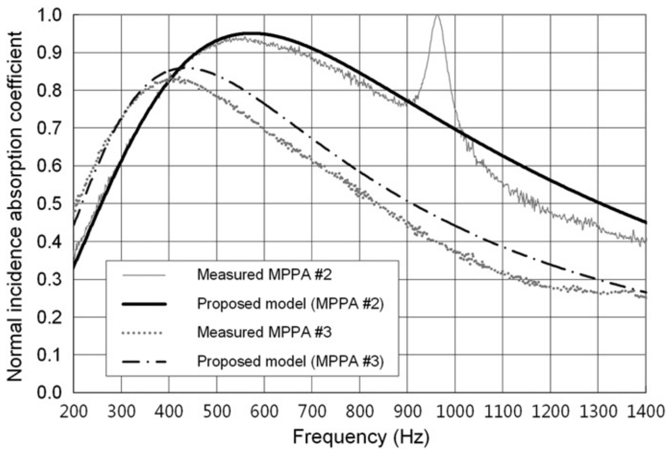

Fig. 8. Normal incidence absorption coefficient comparison of measured and estimated results of MPPA #2 (thickness = 1.0 mm, hole diameter = 0.4 mm, perforation ratio = 1.4%, depth of cavity = 70 mm) and MPPA #3 (thickness = 2.0 mm, other parameters are the same as MPPA #2). The results are estimated at 121 dB.

lower peak of absorption coefficient due to its higher inherent inertance. The predicted results by the current model show good agreement with the measured ones. The results clearly demonstrate that our resistance model can predict the absorption coefficient quite well for MPPAs having different thicknesses. One can observe the sharp absorption peaks at 790 Hz in Fig. 7 and at 970 Hz in Fig. 8. It is noteworthy that Tayong et al. [14] also observed the absorption peak in their measurement results. The peaks are due to the vibration of the panel coupled with air in the back cavity [26,27]: The fundamental frequency of plate vibration coupled with air cavity can be expressed as

$$f_{cp} = \frac{1}{2\pi}\sqrt{\frac{1}{m_p A_{11}}\left(\frac{D_p B_{11}}{L^4} + \frac{\rho_0 c_0^2}{D}\right)}, \tag{14}$$

where m_p is the mass per unit area of the panel, A_{11} and B_{11} are modal constants for the first mode [27], D_p is the bending stiffness of the panel, L is the effective size of the square panel. For clamped boundary condition, the modal constants are

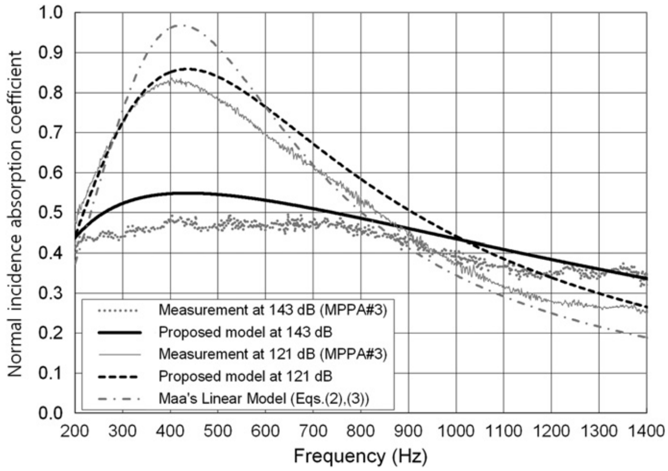

Fig. 9. Normal incidence absorption coefficient comparison of measured and estimated results of MPPA #3 (thickness = 2.0 mm, hole diameter = 0.4 mm, perforation ratio = 1.4%, depth of cavity = 70 mm). The results are estimated at 121 dB and 143 dB.

A_{11} = 2.02 and B_{11} = 2640. It is clear that the second term of Eq. (14) represents the stiffening effect due to the air volume in the back cavity. The estimated frequency by using Eq. (14) for clamped boundary conditions significantly varies according to the effective size of the MPP, which cannot be precisely determined due to the practical clamping condition. We use a few bolts to clamp the MPP around its perimeter (see the fastening hole in Fig. 2). The possible range of L could be from 100 mm to 120 mm if one considers the size of the MPP and the duct. The estimated fundamental resonant frequency from Eq. (14) is in the range of 726 Hz to 1041 Hz. Actually, these are in agreement with the peaks in measurements. In case of 2.0 mm thickness MPP (see measurement results of MPPA #3 in Fig. 8), the resonant frequency is increased due to the thickness ($f_{cp} \sim t^{3/2}$). The estimated frequency is in the range of 2036 Hz to 2930 Hz and the peak is invisible in the figure.

Fig. 9 compares the normal incidence absorption coefficient estimation results and measured ones for the MPPA with lower perforation ratio (MPPA #3, σ = 1.4%). The results illustrate that the current model also predict the absorption coefficient of the MPPA with 1.4% perforation ratio at high incident pressure (143 dB) reasonably well. This example also demonstrates that MPPA with lower perforation ratio loses its absorption characteristics when the incident pressure increases. This indicates that the best design for MPPAs in the low pressure region does not guarantee good performance in high pressure environments such as launcher fairings. The design method for the MPPA for rocket fairings must be different from the conventional ones for the application at low sound pressure levels.

4. MPPA design for launcher fairings

In order to suggest useful guidelines for the design of MPPAs for launcher fairings, we investigate the effect of geometric variables and the incident pressure on the sound absorption characteristics. The absorption coefficient curve can be expressed as three representative measures such as the resonant frequency (where the peak of the absorption coefficient exists), the corresponding peak value of absorption coefficient, and the frequency bandwidth of the absorption coefficient curve. Here, we define the bandwidth as the difference between the upper and lower bound frequencies at which the absorption coefficient curve corresponds to the value of 0.5.

The effects of design variables on the sound absorption ability of MPPA are illustrated in Figs. 10–12. The simulation results at high sound pressure levels are based on the proposed impedance model. The linear impedance model by Maa (Eqs. (2) and (3)) is used to estimate the results at linear pressure regions. In the simulation, we assume that the thickness of the MPP is 1.0 mm and the depth of the cavity is 100 mm. First, we show the resonant frequency variation according to the remaining design variables (the perforation ratio, the hole diameter and the incident pressure level) in Fig. 10. The resonant frequency of MPPA will be lower when the hole diameter becomes larger. The high incident pressure level makes the resonant frequency of MPPA higher. The difference of resonant frequency due to the hole diameter becomes negligible at high incident pressure levels (143 dB). This means that the effect of the hole diameter on the peak frequency of the absorption coefficient curve is not significant for the design of MPPAs for launcher fairings. The effects of design variables on the peak value of absorption coefficient are shown in Fig. 11. When the incident pressure is low, it is clear that MPPAs with a small hole diameter (d = 0.4 mm) and low perforation ratio are good absorbers. MPPAs with a large hole diameter (d = 1.0 mm) and high perforation ratio are poor absorbers in low pressure environments. However, the latter can be a better sound absorber in high pressure environments. Fig. 12 shows the effects of design variables on the frequency bandwidth of MPPA. Note that the minute hole diameter is required to broaden the absorption bandwidth in low pressure environments. This can be readily observed in the figure (see the curves corresponding to "Lin. region"). At high pressure levels, the absorption bandwidth becomes significantly wider and the effect of the hole diameter is negligible for MPPAs

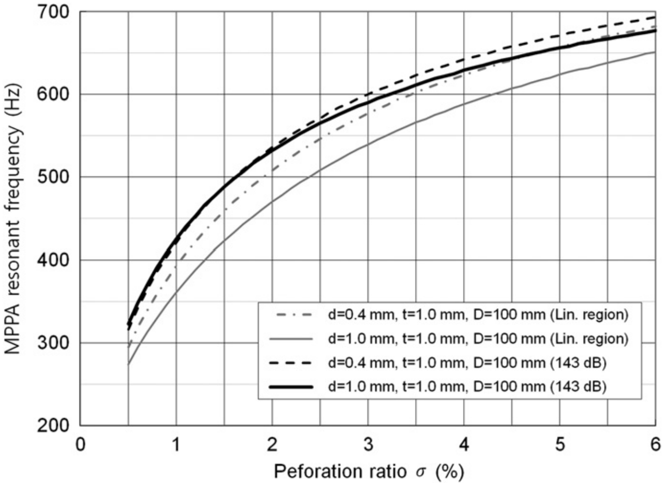

Fig. 10. The effect of design variables on the resonant frequency of the MPPA at both low and high pressure levels.

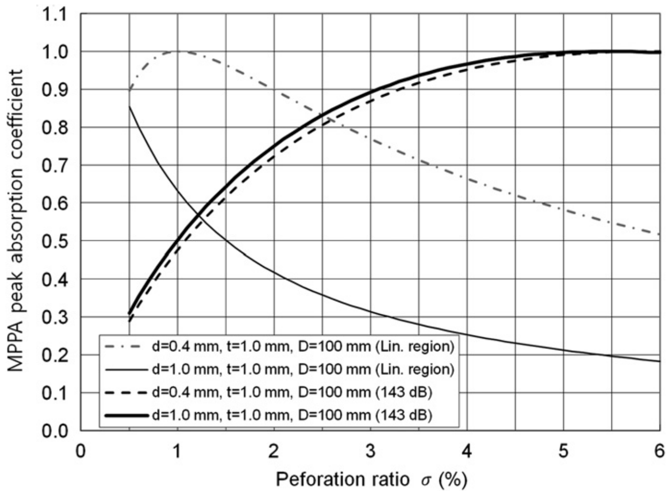

Fig. 11. The effect of design variables on the peak absorption coefficient of the MPPA at both low and high pressure levels.

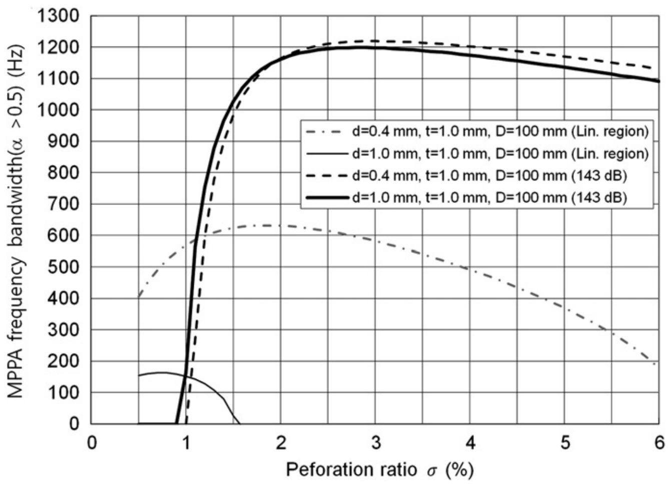

Fig. 12. The effect of design variables on the absorption bandwidth of the MPPA at both low and high pressure levels. The bandwidth is defined as the difference between the upper and lower frequencies at which the absorption coefficient curve corresponds to the value of 0.5.

with a high perforation ratio. This enables us to avoid the use of minute holes to widen the absorption bandwidth, which limits the practical applicability of MPPAs due to its high manufacturing cost. The application of MPPA to launcher fairings seems to be more feasible than the conventional application.

If one considers the above representative measures, then one can readily design a MPPA in a high sound pressure environment. Since the thickness of the MPP and the cavity depth are restricted by the weight and volume requirements, the hole diameter and the perforation ratio as well as the incident pressure could be major design variables. When the thickness of the MPP, the cavity depth and the incident pressure level in the band of interest are given, the optimum values of the hole diameter and the perforation ratio can be determined by the contour plot shown in Fig. 13. Fig. 13(a) shows the design chart of the MPPA at 143 dB SPL. This contour plot is compared with that at very low pressure levels (Fig. 13(b)). The dotted lines indicate the resonant frequency of the MPPA. Note the frequency values on the curve. The thin solid lines represent the peak value of the absorption coefficient. The thick grey solid lines show the bandwidth of the MPPA. Fig. 13(a) clearly indicates that the effect of the hole diameter is negligible at the high pressure level. However, Fig. 13(b) shows that the diameter of the hole governs the absorption characteristics of the MPPA at low pressure levels. Small diameters (below 0.3 mm) are required to design broad band sound absorbers. In addition, the optimum perforation ratio exists according to the diameter of the hole at low pressure levels. For example, one could choose $d=0.2$ mm and $\sigma=4.0\%$ to design a broad band MPPA whose peak frequency and peak absorption is around 600 Hz and 1.0, respectively. However, at high pressure levels, one can choose a wide range of the perforation ratios that makes MPPAs good sound absorbers. We can estimate the absorption characteristics of MPPA #1 (1.0 mm hole diameter and 5.14% perforation ratio) as well. The design chart (Fig. 13(a)) indicates that MPPA #1 will have the normal incidence absorption coefficient curve whose peak absorption coefficient is around 1.0 at a frequency between 600 Hz and 700 Hz (the 700 Hz line is beyond the axis limit and not shown in Fig. 13(a)). The corresponding bandwidth would be about 1100 Hz. One can easily verify this result by observing Fig. 7.

As a practical example, we demonstrate the applicability of MPPAs to mitigate acoustic loadings in payload fairings (PLF). The PLF of NARO (KSLV-I) rocket is chosen as the model PLF. The diameter of the model PLF is 2 m and the length is

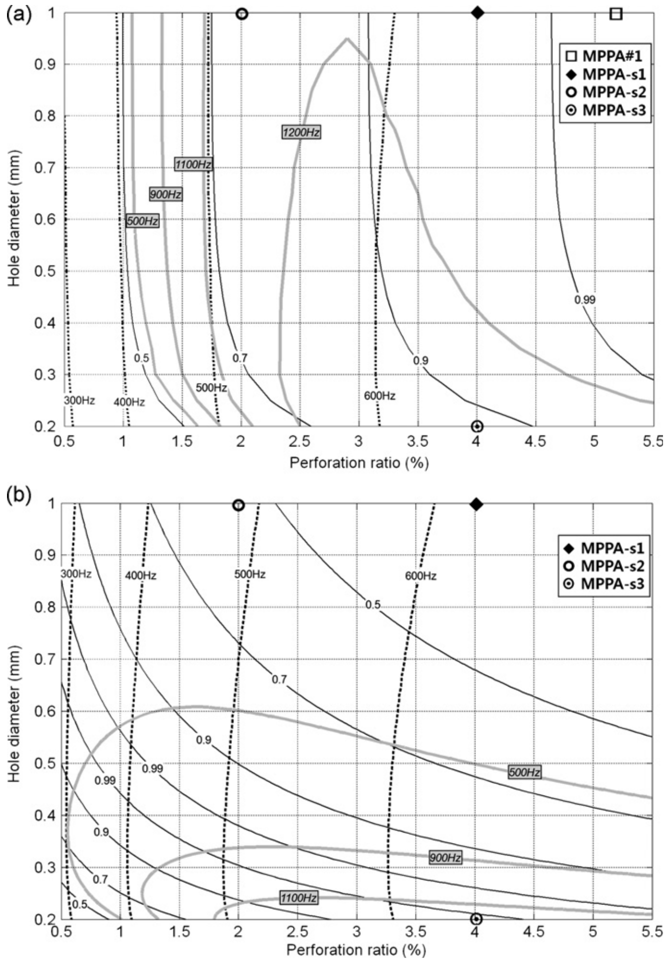

Fig. 13. Design chart for the MPPA (thickness = 1.0 mm, cavity depth = 100 mm): (a) at 137 dB incident pressure level. The corresponding coordinate of the design parameters of MPPA #1 is marked on the plot. (b) In the linear region.

about 5.5 m. We employ the boundary element method to predict acoustic responses inside the PLF. Fig. 14(a) shows the boundary element model used to evaluate noise reduction. In order to calculate acoustic responses up to 800 Hz, we use the very fine boundary element model whose nominal element size is 60 mm: The total number of the boundary element of the model is about 10,000. We assume that a broad band acoustic source is placed at the cone of the PLF, which could excite all acoustic modes efficiently. The pressure responses are calculated at 36 prediction points (on the radius 0.8 m) inside the PLF (see Fig. 14(a)). The commercial boundary element software, LMS Virtual Lab. (ver. 10), is used to calculate the responses. First, we calculate the acoustic responses of the bare fairings (MPPAs are not applied to the PLF in this case). The inherent acoustic damping ratio is assumed to be 1.8%. Next, we consider three different MPPA (denoted by MPPA-s1, MPPA -s2, and MPPA -s3) variants to reduce the acoustic loadings. The geometric parameters of them are shown in Table 2. Corresponding parameters are also marked with symbols in Fig. 13(a) and (b). We apply each MPPA candidate to the cylinder part of the PLF. The application area is determined by considering the practical situation (see Fig. 14(b)). The covering area of the MPPA is 55% of the inner surface area of the PLF. We estimate the normal impedances of the MPPAs at 143 dB by the proposed impedance model (Eqs. (11) and (12)). We import the estimated normal impedance value of each MPPA to the boundary element model, and we consider it as the impedance boundary condition on the surface where the MPPA is applied. Note that the normal impedance fully describes the acoustic characteristics of locally reacting sound absorbers such as MPPAs.

Fig. 15 illustrates the absorption coefficients of the MPPAs at 143 dB (Fig. 15(a)) and at a very low level (Fig. 15(b)). At 143 dB, MPPA-s1 (d=1.0 mm, σ=4.0%) shows the highest absorption coefficient above 250 Hz, but MPPA-s2 (d=1.0 mm,σ=2.0%) shows better absorption below 250 Hz. Two MPPAs of 1.0 mm diameter (MPPA-s1 and MPPA-s2) show very poor sound absorption at low pressure levels (see Fig. 15(b)). MPPA-s3 (d=0.2 mm,σ=4.0%) shows good absorption both at high and low pressure levels. However, the hole diameter is so small, one cannot easily produce the MPPA with 0.2 mm hole diameter by using conventional manufacturing methods. The required non-conventional method would increase the manufacturing cost significantly.

The calculated results are shown in Fig. 16. Fig. 16(a) illustrates the noise reduction (NR) in dB when each of MPPAs is applied to the PLF. Above 200 Hz, MPPA-s1 shows the best performance and the noise reduction reaches 18 dB at

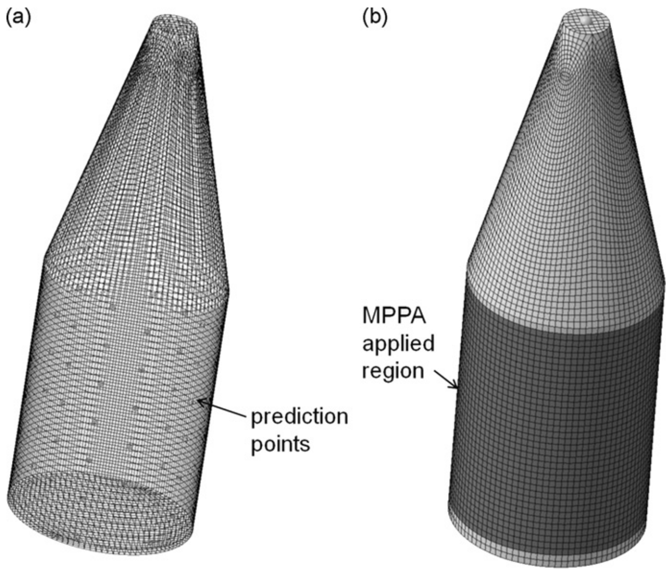

Fig. 14. The boundary element model of the model payload fairing (PLF). (a) Boundary element mesh and 36 prediction points. (b) The region of the boundary element mesh where one of three candidate MPPAs (MPPA-s1, MPPA -s2, and MPPA-s3) is applied as the impedance boundary condition.

Table 2
Geometric parameters of MPPAs for the reduction of noise inside the model PLF.

	Thickness (mm)	Hole diameter (mm)	Perforation ratio (%)	Cavity depth (mm)
MPPA-s1	1.0	1.0	4.0	100
MPPA-s2	1.0	1.0	2.0	100
MPPA-s3	1.0	0.2	4.0	100

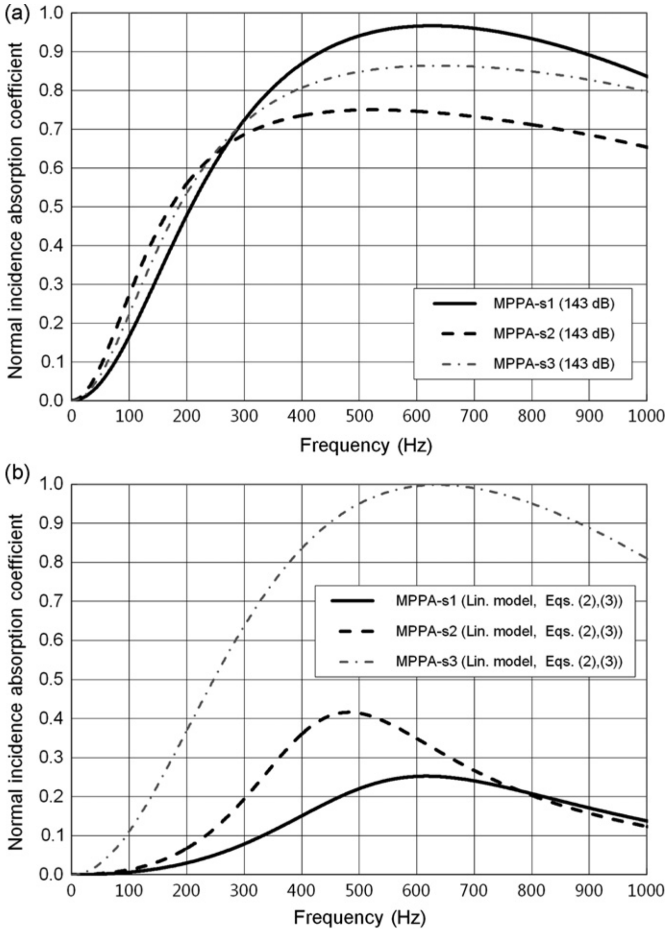

Fig. 15. Normal incidence absorption coefficient curves of three candidate MPPAs (MPPA-s1, MPPA-s2, and MPPA-s3): (a) at 143 dB (b) At low pressure levels where the linear impedance model can be applied. The absorption characteristics of the MPPAs also can be estimated by using the design plots in Fig. 13. The corresponding point of geometric parameters of each MPPA is also marked on Fig. 13.

500 Hz-1/3 octave band. The noise reduction by MPPA-s1 is 3 dB higher than that by MPPA-s3 above 315 Hz band. However, below 200 Hz, MPPA-s2 is better than MPPA-s1 and MPPA-s3. The negative noise reduction at 63 Hz-1/3 octave band is due to the shift of the acoustic resonance of the internal acoustic cavity of the PLF after applying the MPPA. Fig. 16(b) shows the 1/3 octave band spectra of the pressure inside the PLF before and after applying each MPPA. The overall SPL (from 25 Hz- to 630 Hz-1/3 octave bands) of the bare fairing is 145.8 dB. If we apply one of the MPPAs to the PLF, we could reduce the internal SPL by 6.5 dB (MPPA-s1) to 6.9 dB (MPPA-s2, MPPA-s3). It is noteworthy that MPPA-s2 shows good noise reduction in overall sound pressure level. This is due to the characteristic of the pressure spectrum of the bare fairings: Most severe acoustic loading exists in the range of 100 Hz to 300 Hz. These results enable us to choose MPPA-s2 ($d=1.0$ mm,$\sigma=2.0\%$) rather than MPPA-s1 ($d=1.0$ mm, $\sigma=4.0\%$) or MPPA-s3 ($d=0.2$ mm,$\sigma=4.0\%$). This could reduce costs significantly by minimizing the number of holes and permitting large holes.

5. Conclusions

It is concluded that the proposed acoustic resistance model of micro-perforated panel absorbers in a high sound pressure environment can be readily applied to the design of MPPAs for the fairings of launch vehicles. The suggested resistance model considers all geometric parameters of the MPP (the hole diameter, the perforation ratio and the thickness of the panel), and it can explain the resistance variation according to the acoustic excitation. The main control variable can be expressed as the incident pressure in the band of interest, which is practical in the design process. We demonstrated that MPPAs with large hole diameter can be a good sound absorber at high incident pressure. This facilitates the use of MPPAs in a high pressure environment with less manufacturing cost. The practical application of the MPPA to the PLF is demonstrated. It is shown that the MPPA designed by the proposed method could reduce the acoustic loadings inside the model PLF by more than 6 dB.

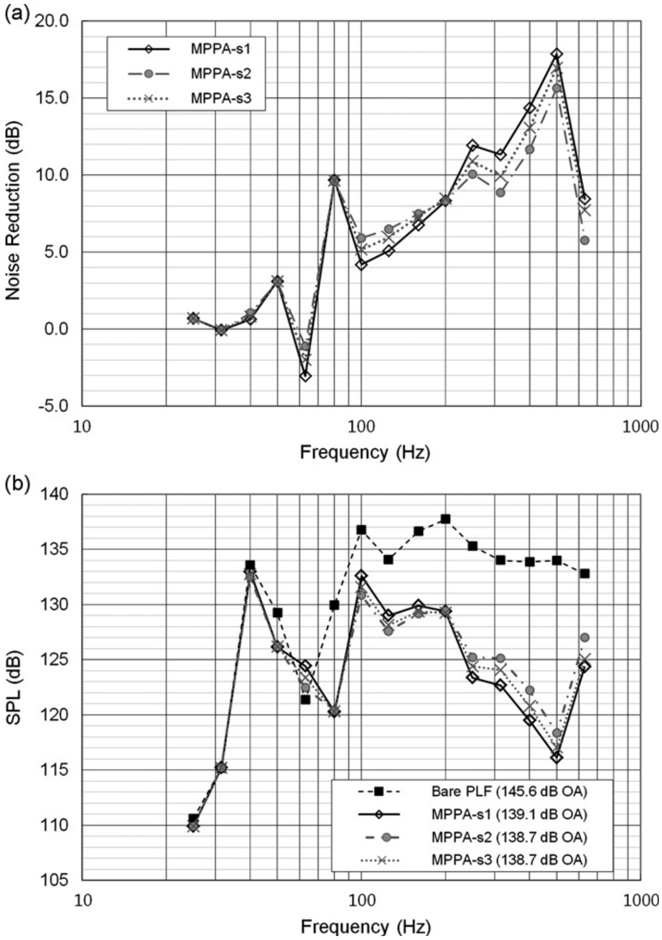

Fig. 16. The results of mitigation of acoustic loads by applying each MPPA to the model PLF. (a) The noise reduction in dB. (b) The 1/3 octave band spectra of the pressure inside the PLF before and after applying each MPPAs to PLF.

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Appendix A. Derivation of the relation between the velocity $\overline{u}_0$ in the orifice and incident pressure $\overline{p}_i$

Acoustic wave in the impedance tube can be regarded as the equivalent acoustic circuit. If one considers the impedance type acoustical circuit [2] where the open-circuit voltage is $2p_i$, the internal resistance is $\rho_0 c_0$, and the load impedance is Z_O, then one can obtain the following relation:

$$\frac{2p_i}{A_1 u_1} = \frac{\rho_0 c_0}{A_1} + \frac{Z_O}{A_1} \tag{A1}$$

where u_1 is the particle velocity (the current in the equivalent circuit) and A_1 is the cross-sectional area of the tube. We assume that an orifice plate is placed on one end of the tube and it is regarded as the load impedance. From the equation of continuity, we can define the particle velocity in the orifice u_0 as

$$u_0 = \frac{u_1}{\sigma}. \tag{A2}$$

where $\sigma = A_0/A_1$ (A_0: the cross-sectional area of the orifice). If one assumes that flow passing the orifice plate is incompressible and laminar, one can apply the momentum equation in the form of Bernoulli's law, then one has the orifice impedance [24] as follows:

$$Z_O = \frac{\rho_0 u_0 (1-\sigma^2)}{\sigma}. \tag{A3}$$

where the mass reactance of the aperture is neglected. From Eqs. (A1), (A2) and (A3), we have

$$2p_i = \rho_0 c_0 \sigma u_0 + \rho_0 u_0^2 (1-\sigma^2). \tag{A4}$$

Note that the variables p_i and u_0 in Eq. (A4) are amplitudes. By using the root-mean-squared variables, Eq. (4) can be rewritten as

$$2\overline{p}_i = \rho_0 c_0 \sigma \overline{u}_0 + \rho_0 \sqrt{2} (\overline{u}_0)^2 (1-\sigma^2). \tag{A5}$$

By solving Eq. (A5) in terms of $\overline{u}_0$, we have the relation between the velocity $\overline{u}_0$ in the hole and incident pressure $\overline{p}_i$ as

$$\frac{\overline{u}_0}{c_0} = \frac{1}{\sqrt{2}} \frac{\sigma}{1-\sigma^2} \left(\sqrt{0.25 + \frac{2\sqrt{2}\overline{p}_i}{\rho_0 c_0^2} \frac{1-\sigma^2}{\sigma^2}} - 0.5 \right). \tag{A6}$$

References

[1] D.-Y. Maa, Potential of microperforated panel absorber, *Journal of the Acoustical Society of America* 104 (5) (1998) 2861–2866.
[2] J. Kang, H.V. Fuchs, Predicting the absorption of open weave textiles and micro-perforated membranes backed by and air space, *Journal of Sound and Vibration* 220 (5) (1999) 905–920.
[3] J. Kang, M.W. Brocklesby, Feasibility of applying micro-perforated absorbers in acoustic window systems, *Applied Acoustics* 66 (6) (2005) 669–689.
[4] J. Tian,Micro perforation for sound absorption and noise reduction, Distinguished Lecture, Inter Noise 2008, Shanghai, China, 2008.
[5] R.E. Kraft, J. Yu and H.W. Kwan, Acoustic treatment design scaling methods, U.S. NASA Report No.CR-1999-209120, 2, 1999.
[6] W.O. Hughes and A.M McNeils, Cassini/Titan IV acoustic blanket development and testing, U.S. NASA Report No. TM 107474, 1996.
[7] T. Bergen, H. Himelblau, D. Kern, Development of acoustic test criteria for the Cassini spacecraft, *Journal of the IEST* 41 (1) (1998) 26–38.
[8] N. Eaton, J. Moran, and P. Roux, Development of lightweight and cost effective acoustic damping system for noise reduction in launcher payload fairings, The 6th International Symposiums on Launcher Technology, Munich, Germany, 2005.
[9] I.U. Borchers, S. T. Laemmlein, P. Bartels, A. Rausch, M. Faust, J.A.F. Coebergh, K. Koeble, Acoustic protection on payload fairings of expendable launch vehicles, U.S. Patent No. 5,670,758, 1997.
[10] S. Lane, R. Richard, S. Kennedy, Fairing noise control using tube-shaped resonators, *Journal of Spacecraft and Rockets* 42 (4) (2005) 640–646.
[11] S. Lane, K. Henderson, A. Williams, Chamber core structures for fairing acoustic mitigation, *Journal of Spacecraft and Rockets* 44 (1) (2007) 156–163.
[12] U. Ingard, S. Labate, Acoustic circulation effects and the nonlinear impedance of orifices, *Journal of the Acoustical Society of America* 22 (2) (1950) 211–218.
[13] D.-Y. Maa, Microperforated panel at high sound intensity, Proceedings of Inter noise 94, Yokohama, Japan, 1994, pp. 1511–1514.
[14] R. Tayong, T. Dupont, P. Leclaire, On the variation of acoustic absorption peak with particle velocity in micro-perforated panels at high level of excitation, *Journal of the Acoustical Society of America* 127 (5) (2010) 2875–2882.
[15] D.A. Bies, O.B. Wilson Jr., Acoustic impedance of a Helmholtz resonator at very high amplitude, *Journal of the Acoustical Society of America* 29 (6) (1957) 711–714.
[16] U. Ingard, H. Ising, Acoustic nonlinearity of an orifice, *Journal of the Acoustical Society of America* 42 (1) (1967) 6–17.
[17] R.L. Panton, A.L. Goldman, Correlation of nonlinear orifice impedance, *Journal of the Acoustical Society of America* 60 (6) (1976) 1390–1396.
[18] R.W. Fox, A.T. McDonald, *Introduction to Fluid Mechanics*, John Wiley & Sons, New York, 1985 Chap. 7, pp. 296–303.
[19] I.B. Crandall, Theory of Vibrating Systems and Sound, *D. Van Norstrand Company* (1926) 230–241.
[20] F. Mechel, I. Ver, Noise and Vibration Control Engineering: Principles and Applications, in: L. Beranek, I. Ver (Eds.), John Wiley and Sons, Inc, New York1992, pp. 216–222.
[21] C. Zwikker, C.W. Kosten, *Sound Absorbing Materials*, Elsevier, New York, 1949 Chap. 7, pp. 132-137.
[22] A.F. Seybert, D.F. Ross, Experimental determination of acoustic properties using a two-microphone random-excitation technique, *Journal of the Acoustical Society of America* 61 (5) (1977) 1362–1370.
[23] U. Ingard, On the theory and design of acoustic resonators, *Journal of the Acoustical Society of America* 25 (6) (1953) 1037–1061.
[24] P.M. Morse, K.U. Ingard, *Theoretical Acoustics*, McGraw-Hill, New York, 1968 Chap. 11, pp. 758–760.
[25] Y. Auregan, M. Pachebat, Measurement of the nonlinear behavior of acoustical rigid porous materials, *Physics of Fluids* 11 (6) (1999) 1342–1345.
[26] Y.Y. Lee, E.W.M. Lee, C.F. Ng, Sound absorption of a finite flexible micro-perforated panel backed by an air cavity, *Journal of Sound and Vibration* 287 (2005) 227–243.
[27] R.D. Ford, M.A. McCormic, Panel sound absorber, *Journal of Sound and Vibration* 10 (1969) 411–423.